

Periodic Noise Words and Exact Markov Moments for an Additive-Noise Hénon Contraction

Route-A finite certificate C123

Abstract

We study an iid two-branch affine Hénon system in exact arithmetic. Strict contraction makes every finite noise word own a unique periodic state. All 126 rooted words through length six yield 23 primitive necklaces. The same source law closes a 15-dimensional Markov operator on polynomials of degree at most four, from which stationary covariance and a non-Gaussian fourth cumulant are recovered. These are finite random-dynamics certificates, not a global Fredholm or arithmetic construction.

1 Frozen additive-noise model

Let $\sigma_n \in \{-1, +1\}$ be iid and uniform, and freeze

$$F_\sigma(x, y) = \left(\frac{x}{2} - \frac{y}{4} + \frac{\sigma}{2}, \frac{x}{4} \right), \quad A = \begin{pmatrix} 1/2 & -1/4 \\ 1/4 & 0 \end{pmatrix}.$$

The common linear part has repeated eigenvalue $1/4$ and determinant $1/16$. The eigenvalues of $A^\top A$ are

$$\frac{3}{16} \pm \frac{\sqrt{2}}{8},$$

both strictly below one. Each branch and every finite composition is therefore a strict affine contraction.

2 Periodic noise-word atlas

For a chronological word $\omega = (\sigma_0, \dots, \sigma_{n-1})$, write $F_\omega = F_{\sigma_{n-1}} \circ \dots \circ F_{\sigma_0}$. Its linear part is A^n , so $I - A^n$ is invertible and F_ω has one exact fixed state. Cyclic rotations change the phase of the same oriented word orbit. We choose the lexicographically least rotation with $- < +$ and remove repeated words.

period n	1	2	3	4	5	6	total
rooted words tested	2	4	8	16	32	64	126
primitive necklaces	2	1	2	3	6	9	23

Every row in the evidence ledger records its canonical word, all rational states, the chosen rooted length- n block probability 2^{-n} , A^n , and composition determinant 16^{-n} . This row value is neither the total mass of its necklace nor an infinite periodic-orbit probability. The cutoff $n \leq 6$ is literal; no all-period completeness is asserted.

3 Degree-four Markov operator

The model itself defines

$$(Pf)(x, y) = \frac{1}{2}f(F_+(x, y)) + \frac{1}{2}f(F_-(x, y)).$$

The 15-dimensional space

$$\mathcal{P}_{\leq 4} = \text{span}\{x^i y^j : i + j \leq 4\}$$

is invariant. In the graded monomial basis, the exact source-owned matrix $K = P|_{\mathcal{P}_{\leq 4}}$ satisfies

$$\text{tr } K = \frac{453}{256}, \quad \det K = 2^{-80}.$$

All sixteen coefficients of $\det(I - zK)$ are stored in the evidence receipt.

Solving $K^\top \mu = \mu$ with $\mu(1) = 1$ gives zero odd moments and

$$\Sigma = \begin{pmatrix} \mu(x^2) & \mu(xy) \\ \mu(xy) & \mu(y^2) \end{pmatrix} = \frac{1}{3375} \begin{pmatrix} 1088 & 128 \\ 128 & 68 \end{pmatrix}, \quad \det \Sigma = \frac{256}{50625}.$$

It obeys $\Sigma = A\Sigma A^\top + \text{diag}(1/4, 0)$ exactly. The fourth cumulant

$$\mu(x^4) - 3\mu(x^2)^2 = -\frac{47789203456}{359401303125}$$

is nonzero, so we do not replace the stationary law by a Gaussian closure. With noise removed, the stable deterministic system has zero stationary covariance.

4 Validation and boundary

From the package directory run

```
python3 code/c123_noise_producer.py
python3 code/c123_noise_checker.py
python3 code/c123_sympy_crosscheck.py
python3 code/c123_replay.py
python3 code/c123_mutation.py
```

The independent checker, fresh symbolic reconstruction, canonical replay, and all 19 hostile mutations pass.

The canonical verdict is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL), overall ROUTE_A_EXPLORATORY. The finite word ledger has no prime-like target correspondence. The degree-four Markov determinant has no target-divisor match or analytic bridge, and no global analytic structure is established. We claim no complete random orbit atlas, global nuclear/Fredholm owner, arithmetic local data, Euler factors, root numbers, automorphy, Hilbert–Pólya operator, or Route B. The firewall is NO_BAD_EULER_OR_ROOT_NUMBER.